



Sprite Audit — The Binding of Isaac 3DS

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Scope: All game entities vs. all sprite assets across 6 texture atlases

Project: binding_of_isaac_3ds/

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Summary Dashboard

Category	Total Entities	✔ Has Sprite	🔧 Placeholder Only	✗ Missing	📦 Unused Atlas Sprite
Player	1 (15 poses)	15	0	0	0
Enemies	6 regular	6	0	0	0
Bosses	5	5	0	0	0
Items	20	20*	0	0	0
Tears (player)	3 color sets	3	0	0	0
Enemy Shots	1 type	0	1	1	0
Effects	5 types	2	0	0	3
Environment	12 tile types	10	0	0	2
Doors	12 variants	12	0	0	0
UI / HUD	8 elements	5	1	0	2
Menu	1	1	0	0	0

***20 items share only 21 unique sprites — 9 items borrow thematically incorrect sprites. See [Items with Mismatched Sprites](#).**

Totals: 8 unused atlas sprites · 2 entities using procedural-only rendering · 9 items need unique sprites

Player Sprites

Atlas: `sprites.t3x ( sheet_sprites ) — include/sprites_atlas.h`

Renderer: `render_player()` (main.c:3306) — full sprite rendering with procedural fallback

Sprite	Atlas Index	Used In	Status
player_down	0	Walk down frame 0 / idle	✓
player_down2	1	Walk down frame 1	✓
player_up	2	Walk up frame 0 / idle	✓
player_up2	3	Walk up frame 1	✓
player_left	4	Walk left frame 0 / idle	✓
player_left2	5	Walk left frame 1	✓
player_right	6	Walk right frame 0 / idle	✓
player_right2	7	Walk right frame 1	✓
player_shoot_down	8	Shooting downward	✓
player_shoot_up	9	Shooting upward	✓
player_shoot_left	10	Shooting left	✓
player_shoot_right	11	Shooting right	✓
player_hurt	12	Damage flinch	✓
player_pickup	13	Item pickup pose	✓
player_death	14	Death animation	✓

Files: gfx_clean/resized/player_*.png (15 files)

Procedural fallback: Circle body + dot eyes (main.c:3367-3373)

Enemies

Regular Enemies (sprites_atlas — static sprites)

Atlas: sprites.t3x (sheet_sprites)

Renderer: render_enemies() (main.c ~2800-3000)

Enemy	Enum	Atlas Sprite	Index	Status
Fly	ENEMY_FLY	enemy_fly	15	✓
Gaper	ENEMY_GAPER	enemy_gaper	16	✓
Pacer	ENEMY_PACER	enemy_pacer	17	✓ (also has animated atlas)
Spider	ENEMY_SPIDER	enemy_spider	18	✓
Clotty	ENEMY_CLOTTY	enemy_clotty	19	✓ (also has animated atlas)
Small Gaper	ENEMY_GAPER_SMALL	enemy_gaper (reused)	16	✓ (scaled down)

Files: gfx_clean/resized/enemy_*.png (5 files)

Procedural fallback: Colored circles/rectangles (main.c render_enemies else-branch)

Animated Enemy Sprites (enemies_atlas)

Atlas: enemies.t3x (sheet_enemies) — include/enemies_atlas.h

Used for: Clotty and Pacer only — animated multi-frame rendering

Sprite Set	Frames	Index Range	Used By
clotty_idle_0..3	4	0-3	Clotty idle animation
clotty_shoot_0..3	4	4-7	Clotty shooting animation
clotty_angry_0..2	3	8-10	Clotty angry state
pacer_walk_00..21	22	11-32	Pacer walk cycle

Files: gfx_clean/resized/enemies/ (33 files)

Bosses

Atlas: sprites.t3x (sheet_sprites) — include/sprites_atlas.h

Renderer: render_enemies() boss branch (main.c ~2900-3000)

Boss	Enum	Atlas Sprite	Index	Status	Notes
Duke of Flies	EN- EMY_BOSS_DUK E	boss_duke_of _flies	20	✓	Spawns flies (which have sprites)
Monstro	EN- EMY_BOSS_MON STRO	boss_monstro	21	✓	Jump shadow is procedural circle
Gemini	EN- EMY_BOSS_GEM INI	boss_gemini	22	✓	Small twin uses same sprite (scaled)
Larry Jr.	EN- EMY_BOSS_LAR RY	boss_larry_j r	23	✓	Segments re- use same sprite
Famine	EN- EMY_BOSS_FAM INE	boss_famine	24	✓	Charge phase uses same sprite

Files: gfx_clean/resized/boss_*.png (5 files)

Procedural fallback: Colored circles with unique colors per boss

Items

Atlas: ui_items.t3x (sheet_ui_items) — include/ui_items_atlas.h

Renderer: Pedestal items via item_sprite_idx() → spr_draw() (main.c:3184-3185)

Mapping: sprite.c:item_sprite_idx() (lines 166-223)

Item	Enum	Maps To Sprite	Atlas Index	Status	Accurate?
Pentagram	ITEM_PENTAGRAM	item_pentagram	8	✓	✓ Correct
The Belt	ITEM_BELT	item_the_belt	9	✓	✓ Correct
Wire Coat Hanger	ITEM_WIRE_COAT	item_technology	18	✓	⚠ Wrong sprite
Lunch	ITEM_LUNCH	item_coin	24	✓	⚠ Wrong sprite
Cupid's Arrow	ITEM_CUPIDS_ARROW	item_moms_knife	19	✓	⚠ Wrong sprite
Spoon Bender	ITEM_SPOON_BENDER	item_spoon_bender	27	✓	✓ Correct
Polyphemus	ITEM_POLYPHEMUS	item_polyphemus	12	✓	✓ Correct
Inner Eye	ITEM_INNER_EYE	item_inner_eye	28	✓	✓ Correct
Spelunker Hat	ITEM_SPELUNKER_HAT	item_bomb	22	✓	⚠ Wrong sprite
Speed Ball	ITEM_SPEED BALL	item_key	23	✓	⚠ Wrong sprite
Magic Mushroom	ITEM_MAGIC_MUSH	item_magic_mushroom	10	✓	✓ Correct
Sacred Heart	ITEM_SACRED_HEART	item_sacred_heart	11	✓	✓ Correct
Spirit Sword	ITEM_SPIRIT_SWORD	item_brimstone	17	✓	⚠ Wrong sprite
Blood of the Martyr	ITEM_BLOOD_OF_MARTYR	item_dead_cat	16	✓	⚠ Wrong sprite
Stigmata	ITEM_STIGMATA	item_cricket_s_head	15	✓	⚠ Wrong sprite
Sad Onion			13	✓	✓ Correct

Item	Enum	Maps To Sprite	Atlas Index	Status	Accurate?
	ITEM_SAD_ONION	item_sad_onion			
Wire Hanger	ITEM_WIRE_HANGER	item_the_pact	20	✓	⚠ Wrong sprite
Growth Hormones	ITEM_GROWTH_HORMONES	item_steven	26	✓	⚠ Borrowed
Jesus Juice	ITEM_JESUS_JUICE	item_steven	26	✓	⚠ Borrowed (duplicate)
The Halo	ITEM_HALO	item_halo	14	✓	✓ Correct

Files: gfx_clean/resized/item_*.png (21 files — all in atlas)

Projectiles & Effects

Atlas: bulletatlas.t3x (sheet_bullets) — include/bulletatlas.h

Atlas: sprites.t3x (sheet_sprites) — effects in sprites atlas

Player Tears ✓

Renderer: render_tears() (main.c:3387-3470)

Tear Type	Sprites	Index Range	Status
Blue (normal)	tear_blue_1..7	0-6	✓ 7 size variants
Red (enemy)	tear_red_1..7	7-13	✓ 7 size variants
Dark (homing)	tear_dark_1..7	14-20	✓ 7 size variants

Tear Effects

Effect	Sprites	Index Range	Status
Tear pop/splash	<code>tear_pop_1..4</code>	21-24	 Used in <code>spawn_tear_pop()</code>
Blood splat (small)	<code>blood_splat_small_1..3</code>	25-27	 Used in <code>spawn_blood_splatter()</code>
Blood splat (medium)	<code>blood_splat_med_1</code>	28	 Used in blood particles
Blood splat (large)	<code>blood_splat_large_1..3</code>	29-31	 Used in blood particles
Blood splat (huge)	<code>blood_splat_huge_1..2</code>	32-33	 Used in blood particles

Enemy Shots

Renderer: `render_enemy_shots()` (main.c:3893-3901) — **procedural only!**

Entity	Current Rendering	Status
<code>EnemyShot</code>	<code>C2D_DrawCircleSolid()</code> — dark red circle + highlight	 Placeholder — no sprite

Red tear sprites (`tear_red_*`) exist in the bullet atlas and could be used.

Unused Projectile/Effect Sprites

Sprite	Atlas	Index Range	Status
<code>tear_trail_1..3</code>	bulletatlas	34-36	 Unused — not referenced in code
<code>projectile_tear</code>	sprites	25	 Unused — bullet atlas tears used instead
<code>effect_blood_1</code>	sprites	26	 Unused — bullet atlas blood splats used instead
<code>effect_blood_2</code>	sprites	27	 Unused — bullet atlas blood splats used instead
<code>effect_explosion</code>	sprites	28	 Unused — no explosion mechanic implemented

Environment & Tiles

Atlas: `environment.t3x` (`sheet_environment`) — `include/environment_atlas.h`

Renderer: `render_room()` (`main.c:3035-3300`)

Floor Tiles

Sprite	Index	Used For	Status
<code>env_floor_clean</code>	0	Normal rooms (floors 0-1)	
<code>env_floor_bloody</code>	1	Boss rooms	
<code>env_floor_cracked</code>	2	Deep floors (≥ 2)	

Wall Tiles

Sprite	Index	Used For	Status
env_wall_top	16	Top wall	
env_wall_bottom	17	Bottom wall	
env_wall_left	18	Left wall	
env_wall_right	19	Right wall	
env_corner_tl	12	Top-left corner	
env_corner_tr	13	Top-right corner	
env_corner_bl	14	Bottom-left corner	
env_corner_br	15	Bottom-right corner	

Doors

Direction	Normal	Treasure	Boss
Top	 env_door_top (6)	 env_door_treasure_top (20)	 env_door_boss_top (24)
Bottom	 env_door_bottom (7)	 env_door_treasure_bottom (21)	 env_door_boss_bottom (25)
Left	 env_door_left (8)	 env_door_treasure_left (22)	 env_door_boss_left (26)
Right	 env_door_right (9)	 env_door_treasure_right (23)	 env_door_boss_right (27)

Obstacles & Objects

Sprite	Index	Used For	Status
env_rock	4	Rock obstacles	
env_trapdoor	5	Floor trapdoor	

Unused Environment Sprites

Sprite	Index	Status	Notes
<code>env_stone_wall</code>	3	 Unused	Generic wall texture — directional walls used instead
<code>env_wall_graffiti</code>	10	 Unused	Decorative wall overlay — never rendered
<code>env_wall_blood</code>	11	 Unused	Decorative wall overlay — never rendered

UI Elements

Atlas: `ui_items.t3x ( sheet_ui_items ) — include/ui_items_atlas.h`

Renderer: `render_hud()` (main.c:2674-2875)

Hearts (HUD)

Sprite	Index	Used In	Status
<code>heart_red_full</code>	0	HUD + pickups	
<code>heart_red_half</code>	1	HUD + pickups	
<code>heart_red_empty</code>	2	HUD empty containers	
<code>heart_soul_full</code>	3	Soul heart pickups	

Unused Heart Sprites

Sprite	Index	Status	Notes
<code>heart_soul_half</code>	4	 Unused	Half soul hearts not implemented in gameplay
<code>heart_black_full</code>	5	 Unused	Black hearts not implemented in gameplay

Other UI

Sprite	Index	Used In	Status
<code>ui_boss_healthbar</code>	6	Boss health bar bg	
<code>ui_item_pickup</code>	7	Item pickup overlay	 Unclear — pedestal rendering is procedural glow

Pedestal Rendering

The item pedestal base is rendered **procedurally** (rectangles + circles at `main.c:3152-3179`). The `ui_item_pickup` sprite exists but the pedestal itself doesn't use it — only the item on top uses `item_sprite_idx()`.

Menu

Atlas: `menu_logo.t3x` (`sheet_menu_logo`) — `include/menu_logo_atlas.h`

Renderer: `render_menu()` (`main.c:2614-2696`)

Sprite	Index	Status	Notes
<code>menu_logo</code>	0		Animated with pulse + bob + glow shadow

Fallback: Text title “The Binding of Isaac” rendered with `C2D_Text` if logo fails to load.

Unused Sprites (In Atlases But Not Referenced in Code)

#	Sprite	Atlas	File Path	Why Unused
1	env_stone_wall	environment	gfx_clean/resized/ env_stone_wall.png	Replaced by directional wall tiles
2	env_wall_graffiti	environment	gfx_clean/resized/ env_wall_graffiti.png	Decorative overlay never implemented
3	env_wall_blood	environment	gfx_clean/resized/ env_wall_blood.png	Decorative overlay never implemented
4	heart_soul_half	ui_items	gfx_clean/resized/ui_hearts/ heart_soul_half.png	Half soul hearts not in gameplay
5	heart_black_full	ui_items	gfx_clean/resized/ui_hearts/ heart_black_full.png	Black hearts not in gameplay
6	projectile_tear	sprites	gfx_clean/resized/projectile_tear.png	Superseded by bullet atlas tear variants
7	effect_blood_1	sprites	gfx_clean/resized/effect_blood_1.png	Superseded by bullet atlas blood splats
8	effect_blood_2	sprites	gfx_clean/resized/effect_blood_2.png	Superseded by bullet atlas blood splats
9	effect_explosion	sprites	gfx_clean/resized/effect_explosion.png	No explosion mechanic in game
10	tear_trail_1..3	bulletatlas	gfx_clean/resized/bullets/	

#	Sprite	Atlas	File Path	Why Unused
			tear_trail_*.png	Trail rendering not implemented

Entities Using Placeholder (Procedural) Graphics

#	Entity	Current Rendering	Suggested Fix
1	Enemy Shots (<code>EnemyShot</code>)	Dark red circle + highlight dot (main.c: 3898-3899)	Use <code>tear_red_*</code> sprites from bullet atlas with smaller scale
2	Item Pedestal base	Procedural rectangles + glow circles (main.c:3152-3179)	Could use <code>ui_item_pickup</code> sprite for pedestal base

Entities with Both Sprite + Procedural Paths (Working Correctly)

Every major entity has a procedural fallback gated by `g_sprites_loaded` . These fallbacks activate only if atlas loading fails:

- **Player:** Circle body + dot eyes → 15 sprite poses
- **All enemies:** Colored circles → individual sprites + animated sheets
- **All bosses:** Colored circles → boss sprites
- **Tears:** Colored circles → 21 sized tear sprites
- **Room (floor/walls/doors):** Colored rectangles → tiled sprites
- **Obstacles:** Brown rectangles → rock sprites
- **Trapdoor:** Concentric circles → trapdoor sprite
- **Hearts (HUD):** Procedural heart shape (`draw_heart()`) → heart sprites
- **Boss health bar:** Green rectangle → `ui_boss_healthbar` sprite
- **Menu logo:** Text fallback → logo sprite

Items with Mismatched/Shared Sprites

These items are implemented in gameplay but display **another item's** sprite on the pedestal:

#	Item (Game Logic)	Displays As	Actual Sprite	Why
1	Wire Coat Hanger	Technology	item_technology	No wire coat hanger sprite exists
2	Lunch	Coin	item_coin	No lunch sprite exists
3	Cupid's Arrow	Mom's Knife	item_moms_knife	No cupid's arrow sprite exists
4	Spelunker Hat	Bomb	item_bomb	No spelunker hat sprite exists
5	Speed Ball	Key	item_key	No speed ball sprite exists
6	Spirit Sword	Brimstone	item_brimstone	No spirit sword sprite exists
7	Blood of the Martyr	Dead Cat	item_dead_cat	No blood of martyr sprite exists
8	Stigmata	Cricket's Head	item_crickets_head	No stigmata sprite exists
9	Wire Hanger	The Pact	item_the_pact	No wire hanger sprite exists
10	Growth Hormones	Steven	item_steven	No growth hormones sprite exists
11	Jesus Juice	Steven	item_steven	No jesus juice sprite exists (shares with #10)

Items with correct, unique sprites (9/20): Pentagram, The Belt, Spoon Bender, Polyphemus, Inner Eye, Magic Mushroom, Sacred Heart, Sad Onion, The Halo



Uploaded Assets Not Yet Integrated

The following cleaned assets exist in `/home/ubuntu/Downloads/` but are **not** yet added to any atlas:

File	Potential Use	Corresponding Atlas Entry
item_belt_clean-Photo-room.png	Redundant — Belt already in atlas	item_the_belt  already exists
item_trophy_clean-Photo-room.png	New — no trophy item in atlas	Could be used for a new item
item_cricket_clean-Photo-room.png	Redundant — Cricket's Head already in atlas	item_crickets_head  already exists
item_cat_clean-Photo-room.png	Redundant — Dead Cat already in atlas	item_dead_cat  already exists
item_knife_clean-Photo-room.png	Redundant — Mom's Knife already in atlas	item_moms_knife  already exists
item_onion_clean-Photo-room.png	Redundant — Sad Onion already in atlas	item_sad_onion  already exists
item_pact_clean-Photo-room.png	Redundant — The Pact already in atlas	item_the_pact  already exists
item_pentagram_clean-Photo-room.png	Redundant — Pentagram already in atlas	item_pentagram  already exists
item_tarot_clean-Photo-room.png	Redundant — Tarot Card already in atlas	item_tarot_card  already exists
item_tech_clean-Photo-room.png	Redundant — Technology already in atlas	item_technology  already exists
item_bomb_clean-Photo-room.png	Redundant	item_bomb  already exists
item_brimstone_clean-Photo-room.png	Redundant	item_brimstone  already exists
item_coin_clean-Photo-room.png	Redundant	item_coin  already exists
item_halo_clean-Photo-room.png	Redundant	item_halo  already exists
item_key_clean-Photo-room.png	Redundant	item_key  already exists
env_*_clean-Photoroom.png (10 files)	Redundant — environment tiles already in atlas	All environment sprites  already exist

File	Potential Use	Corresponding Atlas Entry
<code>ui_*_clean-Photoroom.png</code> (5 files)	Redundant — UI sprites already in atlas	All UI sprites  already exist
<code>projectile_tear_clean-Photoroom.png</code>	Redundant	<code>projectile_tear</code> exists (though unused)

Recommendations

Priority 1 — Quick Wins (No new art needed)

1. **Use red tear sprites for enemy shots** — `render_enemy_shots()` currently draws procedural circles. Replace with `tear_red_*` sprites from the bullet atlas at a smaller scale. This is a ~10 line code change.
2. **Use `ui_item_pickup` for pedestal base** — The sprite exists in the atlas but isn't used. Could replace the procedural pedestal rectangles.
3. **Implement tear trails** — `tear_trail_1..3` sprites exist in the bullet atlas. Could be drawn behind fast-moving or piercing tears (the piercing trail currently reuses the tear sprite itself).

Priority 2 — Decorative Polish (No new art needed)

1. **Use `env_wall_graffiti` and `env_wall_blood`** — Randomly overlay these on wall segments for visual variety.
2. **Implement soul/black heart types** — Sprites for `heart_soul_half` and `heart_black_full` are already in the atlas. Gameplay code just needs to support these heart types.

Priority 3 — New Art Needed

1. **Create unique sprites for 11 mismatched items:**
 - Wire Coat Hanger, Lunch, Cupid's Arrow, Spelunker Hat, Speed Ball
 - Spirit Sword, Blood of the Martyr, Stigmata, Wire Hanger
 - Growth Hormones, Jesus Juice
2. **Use `effect_explosion` sprite** — If bomb items or explosive effects are ever added, this sprite is ready.

Priority 4 — Cleanup

1. **Remove `projectile_tear` from sprites atlas** — It's fully superseded by the bullet atlas tear variants. Removing it saves atlas space.
2. **Remove `effect_blood_1/2` from sprites atlas** — Fully superseded by bullet atlas blood splats.

Atlas Summary

Atlas File	Sheet Handle	Header	Sprites	Size
romfs/ sprites.t3x	sheet_sprites	sprites_atlas.h	29	Player(15) + Enemy(5) + Boss(5) + Projectile/Effect(4)
romfs/environ- ment.t3x	sheet_environment	environ- ment_atlas.h	28	Floor(3) + Walls(8) + Doors(12) + Rock + Trapdoor + Decor(3)
romfs/ ui_items.t3x	sheet_ui_items	ui_items_atlas. h	29	Hearts(6) + UI(2) + Items(21)
romfs/bulletat- las.t3x	sheet_bullets	bulletatlas.h	37	Tears(21) + Pop(4) + Blood(9) + Trail(3)
romfs/en- emies.t3x	sheet_enemies	enemies_atlas.h	33	Clotty(11) + Pacer(22)
romfs/ menu_logo.t3x	sheet_menu_logo	menu_logo_atlas .h	1	Menu logo

Total atlas sprites: 157

Actually used in rendering code: 147

Unused: 10 sprites across 3 atlases